Block seminar "Purity of the Brauer groups: Gabber's theorem"

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October 17, 2026

Outline

1. Brauer-Azumaya groups
2. Brauer-Grothendieck groups
3. Comparing two groups

Azumaya algebras

Definition

X scheme, $\mathcal{A}$ an $\mathcal{O}_X$-algebra. We say that $\mathcal{A}$ is an *Azumaya algebra* if there exists an étale covering $V_i \to U$ such that $\mathcal{A}|_{V_i} \simeq M_n(\mathcal{O}_{V_i})$.

Definition

Azumaya algebras $\mathcal{A}$ and $\mathcal{B}$ over X are *equivalent* if

$$\mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{E}nd_{\mathcal{O}_X}(\mathcal{F}) \simeq \mathcal{B} \otimes_{\mathcal{O}_X} \mathcal{E}nd_{\mathcal{O}_X}(\mathcal{G})$$

for some finite locally free sheaves $\mathcal{F}, \mathcal{G}$ over X.

Brauer-Azumaya groups

Definition

The *Brauer-Azumaya group*

$\mathrm{Br}_{\mathrm{Az}}(X) = (\text{equivalence classes of Azumaya algebras over } X, \otimes_{\mathcal{O}_X})$.

Brauer-Grothendieck group

Definition

The *Brauer-Grothendieck group*

$$\mathrm{Br}(X) := \mathrm{H}^2_{\text{Ét}}(X, \mathbb{G}_m)$$

Gabber's theorem

Gabber's theorem

There is a natural injection

$$\mathrm{Br}_{\mathrm{Az}}(X) \to \mathrm{Br}(X)_{\mathrm{tors}}$$

for any scheme X. The map is an isomorphism if X admits an ample invertible sheaf L.

Some preliminaries

- A *prestack* is a functor $\mathrm{Sch}/S \to \mathrm{Grpd}$. A *stack* is a prestack F that satisfies étale descent: For any étale covering $Y \to X$,

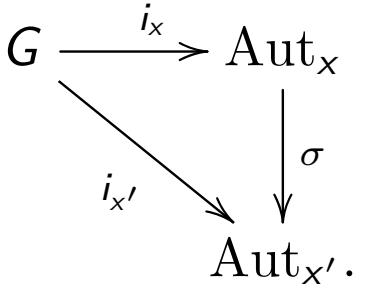
$$F(X) \xrightarrow{\simeq} \lim(F(Y) \rightrightarrows F(Y \times_X Y) \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; \cdots).$$

- A G-gerbe is a stack $F : \mathrm{Sch}/S \to \mathrm{Grpd}$ consists of

$$i_x : G \xrightarrow{\simeq} \mathrm{Aut}_x$$

for any pair $(X \in \mathrm{Sch}/S, x \in F(X))$ which satisfy:

1. Every X admits a cover $\{X_i \to X\}$ such that $F(X_i) \neq \emptyset$.
2. For any $x, x' \in F(X)$, there exists a covering $\{X_i \to X\}$ such that $x|_{X_i} \simeq x'|_{X_i}$.
3. For any $\sigma : x \to x'$,



Some preliminaries

By [Gir71, Thm. IV.3.4.2 (i)], there exists a canonical isomorphism

$$\{\text{isomorphism classes of } G\text{-gerbes}\} \xrightarrow{\simeq} \mathrm{Br}(X).$$

Gabber's theorem

de Jong's proof.

$$\mathrm{Br}(X) \simeq \{\text{isomorphism classes of } \mathbb{G}_m\text{-gerbes}\}, \alpha \mapsto \mathfrak{X}_\alpha.$$

Construct a correspondence

$$(\text{Azumaya algebra } \mathcal{A}) \mapsto \mathfrak{X}(\mathcal{A}) \in \mathrm{Br}(X).$$

Given three stacks:

- $\mathrm{Vb}_X : (T \in \mathrm{Sch}_{/X}) \mapsto (\mathrm{Vb}^{\simeq}_T \in \mathrm{Grpd})$;
- $\mathcal{A} : (T \in \mathrm{Sch}_{/X}) \mapsto ((\mathcal{A}_T, \mathrm{id}_{\mathcal{A}_T}) \in \mathrm{Grpd})$;
- $\mathrm{Alg}_X : (T \in \mathrm{Sch}_{/X}) \mapsto \mathrm{Alg}^{\simeq}_T \in \mathrm{Grpd})$; with morphisms $\mathrm{Vb}_X \xrightarrow{\mathcal{M} \mapsto \mathcal{E}nd(\mathcal{M})} \mathrm{Alg}_X, \mathcal{A} \to \mathrm{Alg}_X$, the homotopy fibre product

$$\mathfrak{X}(\mathcal{A}) := \mathrm{Vb}_X \times^h_{\mathrm{Alg}_X} \mathcal{A} \text{ is a stack.}$$

Gabber's theorem

de Jong's proof.

$\mathfrak{X}(\mathcal{A})$ is a $\mathbb{G}_m$-gerbe:

- Any $T \in \mathrm{Sch}_{/X}$ admits an (étale) covering $\{T_i \to T\}$ such that $\mathfrak{X}(\mathcal{A})(T_i) \neq \emptyset$.
- Any two objects of in $\mathfrak{X}(\mathcal{A})(T)$ are locally isomorphic: Locally,

$$\mathfrak{X}(A)(T) = \{\mathcal{O}_T^n, \mathcal{E}nd(\mathcal{O}_T^n) \xrightarrow{\simeq} \mathcal{E}nd(\mathcal{O}_T^n), \mathcal{E}nd(\mathcal{O}_T^n)\}.$$

[Stacks project, Tag0A2K] $\to$ such isomorphism is given by an isomorphism $\mathcal{O}_T^n \xrightarrow{\simeq} \mathcal{O}_T^n$ and

$$\begin{array}{ccc} \mathcal{O}_T^n & \longrightarrow & \mathcal{O}_T^n \\ \simeq\downarrow & & \downarrow\simeq \\ \mathcal{O}_T^n & \longrightarrow & \mathcal{O}_T^n. \end{array}$$

Gabber's theorem

Equivalently,

$$\begin{array}{ccc} \mathcal{E}nd(\mathcal{O}_T^n) & \xrightarrow{\text{conjugation}} & \mathcal{E}nd(\mathcal{O}_T^n). \\ \Big\downarrow{\scriptstyle \text{conjugation}} & \nearrow_{\mathrm{id}_{\mathcal{O}_T^n}} & \\ \mathcal{E}nd(\mathcal{O}_T^n) & & \end{array}$$

- There are compatible isomorphisms induces by $(\mathcal{M}, \mathcal{E}nd(\mathcal{M}) \simeq \mathcal{A}_T, \mathcal{A}_T) \xrightarrow{\simeq} (\mathcal{M}', \mathcal{E}nd(\mathcal{M}') \simeq \mathcal{A}_T, \mathcal{A}_T)$:

$$\begin{array}{ccc} \mathbb{G}_m & \longrightarrow & \underline{\mathrm{Aut}}_{(T, \mathcal{E}nd(\mathcal{M}) \simeq \mathcal{A}_T, \mathcal{A}_T)} \\ & \searrow & \Big\downarrow \\ & & \underline{\mathrm{Aut}}_{(T, \mathcal{E}nd(\mathcal{M}') \simeq \mathcal{A}_T, \mathcal{A}_T)}. \end{array}$$

Gabber's theorem

- Injectivity: $\mathfrak{X}(A)$ is trivial when $\mathfrak{X}(\mathcal{A})(X) \neq \emptyset$, i.e. $\mathcal{E}nd(\mathcal{M}) \simeq \mathcal{A}$.
- Surjectivity: By [Lie08, Proposition 3.1.2.1(i)],

$$\mathfrak{X} \simeq \mathfrak{X}(\mathcal{A})$$

when $\mathfrak{X}$ admits a finite locally free 1-twisted $\mathcal{O}_{\mathfrak{X}}$-module $\mathcal{M}$.

$$(\text{1-twisted sheaves on } \mathfrak{X}_\alpha) \simeq (\alpha\text{-twisted sheaves on } X_{\text{Ét}}).$$

Some preliminaries

Definition

Given $n \in \mathbb{Z}_{\geq 0}, \mathcal{F} \in \mathrm{QCoh}(\mathfrak{X})$, with $\mathbb{G}_m$-gerbe $\mathfrak{X}$, we say that $\mathcal{F}$ is n-twisted if for any pair (T, x), $\mathrm{Aut}_x \simeq \mathbb{G}_{m,x}$ acts on $\mathcal{F}(T, x)$ via the character $t \mapsto t^n$.

Definition

A. Caldararu: Assume α is represented by Cech cocyle

$$\alpha_{ijk} \in \Gamma(U_{ijk}, \mathbb{G}_m),$$

where $\{U_i \to X\}$ is an étale covering. An α-twisted sheaf is given by $\{\mathcal{M}_i \in \mathrm{QCoh}(U_i), \phi_{ij} : \mathcal{M}_j|_{U_{ij}} \to \mathcal{M}_i|_{U_{ij}}\}$ such that $\phi_{ij} \circ \phi_{jk} = \alpha_{ijk}\phi_{ik}$.

Gabber's theorem

By absolute noetherian approximation,

$$(X, L) \simeq \varprojlim_i (X_i, L_i)$$

where X_i separated, of finite type over $\mathbb{Z}$, L_i is ample, $X_i \to X_j$ is affine. Hence $\mathrm{Br}(X) \simeq \varinjlim \mathrm{Br}(X_i)$ and we may assume X is of finite type over $\mathbb{Z}$.

Gabber's theorem

Pick any $\alpha = [\mathfrak{X}] \in \mathrm{Br}(X)_{\mathrm{tors}}$ with $n\alpha = 0$.
(H_c) For any finite set of closed points $T \subseteq X$, after a finite flat ring extension $R \to R'$, there exists coherent α-twisted sheaf $\mathcal{F}$ which is flat on T such that

$$\mathrm{codim}_X(\mathrm{Sing}(\mathcal{F})) \geq c.$$

$$(H_1) \to (H_2) \to \cdots \to (H_{\dim(X)+1}) \to (\mathrm{Sing}(\mathcal{F}) = \emptyset).$$

Gabber's theorem

- Proof of (H_1) : There exists $s \in \mathrm{H}^0(X, L^{\otimes m})$ s.t X_s is affine, $\mathcal{F}_s$ finite locally free α-twisted sheaf on X_s, and $T \subseteq X_s$. $j : \mathfrak{X}_s \to \mathfrak{X}$ open immersion.

  $$\mathcal{F}_s \leftrightarrow \text{1-twisted finite locally free sheaf } \mathfrak{F}_s \text{ on } \mathfrak{X}_s.$$

  There exist coherent 1-twisted subsheaves $\mathfrak{F}_i$ of $\mathfrak{F}_s$ s.t

  $$j_*\mathfrak{F}_s = \varinjlim_{\text{filtered}} \mathfrak{F}_i.$$

  Hence

  $$\mathfrak{F}_s = j^* j_* \mathfrak{F}_s = \varinjlim j^* \mathfrak{F}_i.$$

Gabber's theorem

There exists $\mathfrak{F} = \mathfrak{F}_i$ satisfying $j^*\mathfrak{F} = \mathfrak{F}_s$. Assume (H_c).
Step 1. After a finite flat $R \to R'$, there exist α-twisted sheaves $\mathcal{F}_0, \ldots, \mathcal{F}_n$ and finite sets of closed points $S_0, \ldots, S_n$ satisfying:

1. $\mathcal{F}_i$ is flat on T;
2. $\mathcal{F}_i$ has constant positive rank on $X - \mathrm{Sing}(\mathcal{F}_i)$;
3. $\mathrm{codim}_X(\mathrm{Sing}(\mathcal{F}_i)) \geq c$, and each irreducible component of codim c contains a point in S_i;
4. $\mathcal{F}_j$ is flat on S_i if $i \neq j$.

Proof of Step 1. There exists $\mathcal{F}_0$ by (H_c) with

$$S_0 = \{1 \text{ closed point in each irr. comp. of codim } c \text{ of } \mathrm{Sing}(\mathcal{F}_0)\}.$$

Set $T_0 = T \cup S_0$. There exists $\mathcal{F}_1$ by (H_c).

Gabber's theorem

There exist closed points in each irreducible component of codim c in $\mathrm{Sing}(\mathcal{F}_1)$ not contained in $\mathrm{Sing}(\mathcal{F}_0)$. This gives S_1 and set $T_1 = T_0 \cup S_1$. Repeating this process gives desired $\mathcal{F}_i$ and S_i.
Step 2. We can assume that $\mathcal{F}_i$ are of the same large rank r on $X - \mathrm{Sing}(\mathcal{F}_i)$. Set

$$G_1 = (\mathcal{F}_0 \oplus \cdots \oplus \mathcal{F}_n)^{\oplus r^n}, \ G_2 = \mathcal{F}_0 \otimes \cdots \otimes \mathcal{F}_n,$$

$$H = \mathcal{H}\mathrm{om}(G_1, G_2).$$

We have G_1, G_2 are coherent α-twisted, and H is coherent $\mathcal{O}_X$-module. Set

$$\psi \in \Gamma(X, H \otimes L^{\otimes N}), \ \mathcal{F} = \ker(\psi), U = X - \bigcup_{i=0^n} \mathrm{Sing}(\mathcal{F}_i).$$

Gabber's theorem

Claim

Assume that

1. $$\psi_{\overline{x}} : G_1 \otimes \kappa(\overline{x}) \to G_2 \otimes L^{\otimes N} \otimes \kappa(\overline{x})$$
   is surjective for every $\overline{x} = \mathrm{Spec}(\kappa(\overline{x})) \in U$.
2. For $i = 0, \ldots, n$ and any $s \in S_i$,
   $$\mathcal{F}_i^{\oplus r^n} \otimes \kappa(s) \to G_1 \otimes \kappa(s) \to G_2 \otimes L^{\otimes N} \otimes \kappa(s)$$
   is an isomorphism and
   $$\left(\bigoplus_{j \neq i} \mathcal{F}_j\right)^{\oplus r^n} \otimes \kappa(s) \to G_1 \otimes \kappa(s) \to G_2 \otimes L^{\otimes N} \otimes \kappa(s)$$
   is 0.

Gabber's theorem

Claim

Then $\mathcal{F}$ is an α-twisted sheaf with $\mathrm{Sing}(\mathcal{F}) \subseteq \cup_{i=0}^{n} \mathrm{Sing}(\mathcal{F}_i)$ which is disjoint from $S = \cup_{i=0}^{n} S_i$. In particular,

$$\mathrm{codim}_X(\mathrm{Sing}(\mathcal{F})) \geq c + 1.$$

Proof of the claim

(a) implies that $\mathcal{F}|_U$ is finite locally free.
Pick $x \in S$. We have

$$\mathrm{Br}(\mathcal{O}^h_{X,x}) \simeq \mathrm{Br}(\kappa(x)) = 0.$$

Since $\mathrm{Br}(\mathcal{O}^h_{X,x}) = \varinjlim_{(B_i,\mathfrak{q}_i)} \mathrm{Br}(B_i)$, there exists a finite étale extension $\mathcal{O}_{X,x} \hookrightarrow B$ such that $\alpha|_B = 0$.

Gabber's theorem

Proof of the claim

Hence there exists a lifting $\mathrm{Spec}(B) \to \mathfrak{X}$ such that $\mathfrak{F}_i|_B \simeq \tilde{M}_i$ where M_i is a finite B-module. For $x \in S_i$ for some i, we arrange that M_j is free of rank r.

$$H \otimes B = \mathrm{Hom}_B(M^{\oplus r^n}, M^{\oplus r^n}) \oplus \mathrm{Hom}_B(B^{\oplus n r^{n+1}}, M^{\oplus r^n}).$$

Write $\psi \otimes B = \psi_1 \oplus \psi_2$. Condition (b) implies ψ_1 is an isomorphism and $\psi_2 = 0$. Hence

$$H \otimes B = B^{\oplus n r^{n+1}} \subseteq G_1 \otimes B.$$

Thus $\mathcal{F}$ is flat at x.

Gabber's theorem

Step 3. We now show that when $N >> 0$, there exists such a section ψ in Step 2. Set $\psi_s = \mathrm{id} \oplus 0$. For $u \in U$

$$C_u = \{A \in H_u \mid \mathrm{rank}(A) < r^{n+1}\}.$$

Claim. Suppose that

- For each $u \in U$, a closed homogeneous subset C_u of $H_u = H \otimes \kappa(u)$ of codim $> \dim(X) + 1$;
- A finite set of closed points $S \subseteq X - U$ and ψ_s for each $s \in S$.

Then there exist $N > 0$, finite flat $R \to R'$, and $\psi \in \Gamma(X_{R'}, H \otimes L^{\otimes N})$ such that $\psi_u \notin C_u \otimes L_u^{\otimes N}$ for $u \in U_{R'}, \psi_{s'} = c\psi_s$ for some $c \neq 0$, and each closed point $s' \in X_{R',s}$.

Gabber's theorem

Proof of the claim

For N sufficiently large: There exists surjection

$$\mathcal{O}_X^I \to \mathcal{I}_S \otimes H \otimes L^{\otimes N}, 1_i \mapsto \Psi_i;$$

$$\Psi_s \in \Gamma(X, \mathcal{I}_{S-\{s\}} \otimes H \otimes L^{\otimes N})$$

whose value at s is ψ_s. Set

$$\mathbb{A} = \mathrm{Spec}(R[x_i, y_s]_{i\in I, s\in S}),$$

$$\Psi = \sum_{i\in I} x_i \Psi_i + \sum_{s\in S} y_s \Psi_s \text{ a section of } H \otimes L^{\otimes N}|_{\mathbb{A}\times X};$$

$$Z = \{(a, u) \in \mathbb{A} \times U \mid \Psi_{a,u} \in C_u\}.$$

Gabber's theorem

Proof of the claim

Each fiber of the projection $Z \to U$

$$\leq |I| + |S| - \mathrm{codim}_{H_u}(C_u).$$

Hence $\dim(Z) < |I| + |S| - 1 = \dim(\mathbb{A}) - 2$. Set $Z' =$ closure of Z in $\mathbb{A}$, then $\mathrm{codim} Z' \geq 2$.
For each $s \in S$, set

$$Z_s = V((X \to \mathrm{Spec}(R))(s), y_s) \subseteq \mathbb{A}$$

Surjective morphism

$$\mathbb{A} - (Z \cup \bigcup_{s \in S} Z_s) \to \mathrm{Spec}(R).$$

Gabber's theorem

We conclude by Rumely's local to global principle: Irreducible $V \to \mathrm{Spec}(R)$ separated, of finite type admits a point in $\overline{R}$ if the structure map is surjective and the generic fiber is geometrically irreducible.

References



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